EXPERIMENT 13

Perform affine transformation on the given image using Python and OpenCV.

Aim

Perform affine transformation on the given image using Python and OpenCV.

Algorithm

• Import OpenCV and NumPy.

• Read the input image.

• Define three source points and three destination points.

• Calculate the affine transformation matrix.

• Apply the transformation and save the image.

Code

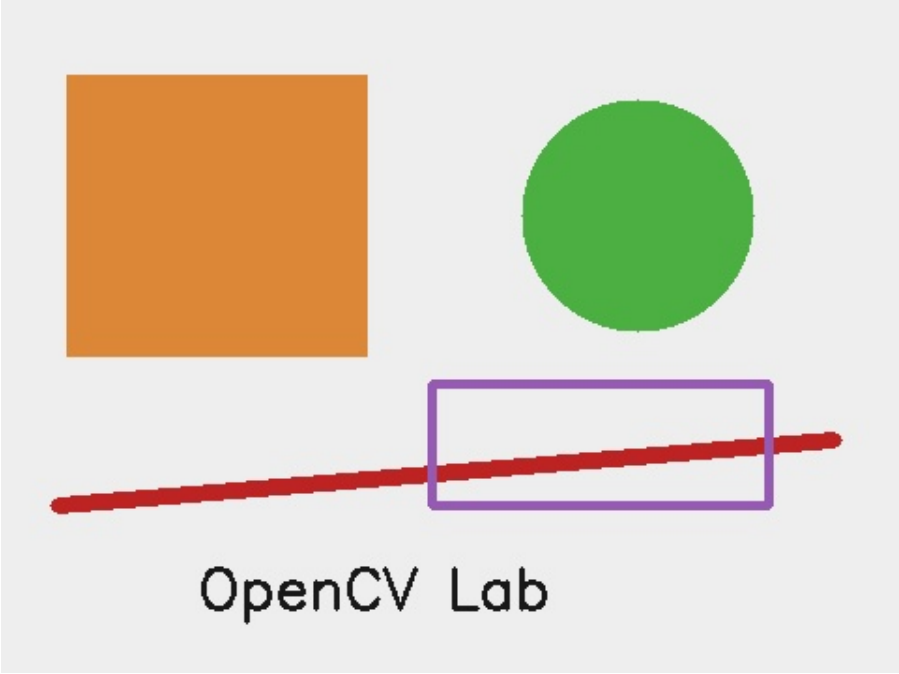
```
import cv2
import numpy as np

img = cv2.imread("images/input.jpg")
if img is None:
    raise FileNotFoundError("images/input.jpg not found")
h, w = img.shape[:2]
src = np.float32([[0, 0], [w - 1, 0], [0, h - 1]])
dst = np.float32([[40, 30], [w - 50, 50], [30, h - 40]])
matrix = cv2.getAffineTransform(src, dst)
result = cv2.warpAffine(img, matrix, (w, h))
cv2.imwrite("output_images/experiment13_affine.png", result)
print("Saved output_images/experiment13_affine.png")
```

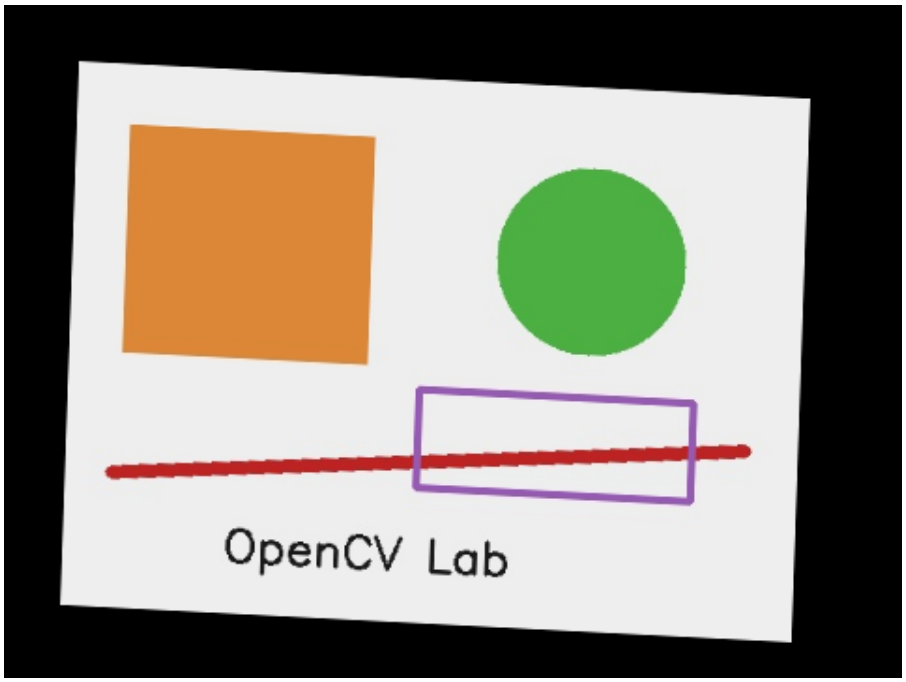
Input Image:

Output Image:

Input Image



Output Image


OpenCV Lab